

Android TV Player/Viewer Validation + Missing Readers — Design

Revision: 1 **Last modified:** 2026-06-29T16:10:00Z **Authority:** Operator mandate (2026-06-29): “physical proofs with hard evidence that all players for all supported content types work and its UI/UX controls ... books, comics, images ... fully tested, validated and verified ... no false results or bluff.” Constitution §11.4.107 (AV liveness battery), §11.4.117 (pixel-oracle), §11.4.143 (real-user journey), §11.4.136 (real-content playback), §11.4.158/§11.4.159 (window-scoped recording + read-the-screen), §11.4.170 (host-side rendered-UI proof), §11.4.52 (autonomous vs operator-attended), §11.4.25 (coverage ledger).

Scope (operator-approved)

Platform: **Android TV only** (connected device MIBOX4 @192.168.0.214:5555). Evidence bar: **full §11.4.107 battery (strictest)**. Decision: **test what exists AND build the missing readers**, then validate them too.

Ground truth (established empirically, §11.4.6 — not assumed)

- **One real player on TV:** `ui/screens/player/MediaPlayerScreen.kt` = `ExoPlayer (Media3)` + custom VLC-style D-pad overlay (play/pause, seek $\pm$, speed presets, back). **Video-focused.** Resolves the stream via `GET /api/v1/entities/:id/stream (stream_url field)`, with auth.
- **Routing:** `MediaDetailScreen.isContainerType` $\rightarrow$ {`tv_show`, `tv_season`, `music_album`} browse children; every other leaf type $\rightarrow$ `onNavigateToPlayer` (the video player).
- **NO dedicated readers exist** for book / comic / image on TV (no Reader/Viewer screens). Selecting those leaf types currently routes to the video player, which cannot render them.
- **Real content present** (catalog DB): 3172 mkv + 2792 mp4 + 672 avi video, 47871 mp3 audio, 28812 jpg + 1623 png images, 2192 cbr + 1230 cbz comics, 93 pdf books.

- **Building blocks:** Coil (coil-compose 2.5.0 + coil-svg) already a dependency. Raw file streaming endpoints exist (/api/v1/entities/:id/stream, /stream/:id). NO paged comic/book reading API yet (only cover/image-proxy).

Phases (each independently shippable)

Phase 1 — Validate what EXISTS, with captured evidence (no new code)

Establish the honest baseline + prove the video/audio players. 1. **Video playback (movie + TV episode)** — the deep case. Drive the real journey (launch → login via §11.4.117 pixel-oracle → browse → select a real mp4/mkv title → Play) and apply the §11.4.107 battery: window-scoped MP4 screen-recording (§11.4.159), freeze-detection (adjacent-frame perceptual-hash distance > threshold = advancing), an independent frame-advance counter from `dumpsys media.metrics / SurfaceFlinger`, not-stale check, measured-fps/decoder-health (dropped-buffer budget per §11.4.136), then exercise EVERY UI/UX control with captured proof: pause (frames stop + counter flat), resume (advance), seek (position jumps), speed change, back (returns to detail). Self-validated analyzer golden-good/golden-bad per §11.4.107(10). 2. **Audio (mp3)** — play a real track; capture player state + on-device audio-session evidence (`dumpsys audio active track / AudioTrack`). Honest boundary: true acoustic output has no clean on-device oracle (§11.4.107 honest-gap) — capture what is provable, mark the rest operator-attended. 3. **Books / comics / images — empirical baseline:** drive the app to a real comic, book, and image item; capture exactly what happens today (video-player error? blank? fallback?). Record the truthful current verdict (expected: NOT-IMPLEMENTED). This is the §11.4.6 baseline that justifies Phase 2 — never a faked pass.

Phase 2 — Build the missing readers (TDD + host-side render proof §11.4.170)

Each viewer is an independent unit (own screen, own route, own tests). Ordered simplest-first: 1. **Image viewer** (jpg/png/webp) — Compose screen using Coil AsyncImage over the streamed file URL; D-pad: next/prev within a directory, zoom, back. Simplest (Coil already present). 2. **Comic reader** (cbz/cbr) — paged image viewer. Needs an **API addition**: an endpoint that lists/extracts pages from a cbz (zip) / cbr (rar) archive (e.g. GET /api/v1/entities/:id/pages + /pages/:n). TV screen: paged D-pad navigation (next/prev page, first/last, back), page-count indicator. 3. **Book/PDF reader** (pdf) — render pages (the API already links go-fitz/mupdf per the build deps); page nav. epub deferred unless trivial. Routing change in MediaDetailScreen: dispatch leaf type →

{image_viewer | comic_reader | book_reader | media_player} by media_type/extension instead of always the video player. Each: §11.4.43 TDD (unit + the §11.4.170 host-side rendered-pixel proof: the Compose screen rendered to PNG on the host via Roborazzi/Paparazzi, golden image-diff + OCR/vision oracle that the page content is legible, light+dark).

Phase 3 — Validate the new readers on-device (full §11.4.107 battery)

Drive each new reader on the MIBOX4 with a real comic/book/image: §11.4.158 read-the-screen (OCR confirms a real page rendered, not blank/error), §11.4.117 pixel-oracle navigation, exercise every control (page next/prev, zoom, back), window-scoped MP4 recording, self-validated analyzer.

Deliverables

- docs/qa/androidtv-players-<run-id>/ — per-content-type evidence (recordings, frame analyses, dumsys captures, screenshots), a **coverage ledger** (§11.4.25): content-type × player/viewer × platform × evidence-path × verdict {PASS | operator-attended | NOT-IMPLEMENTED→built→PASS}.
- New TV reader screens + routing (Phase 2), their unit + host-render tests, the comic-pages API.
- All TDD RED→GREEN, go vet/lint clean, pushed to all 6 remotes; HelixQA bank entries per §11.4.58 layer-4.

Honest boundaries (§11.4.6)

- Only Android TV this effort; Android-phone/desktop/web are separate (their harnesses unverified).
- Acoustic audio output + true photon-FPS at the panel have no clean on-device software oracle — flagged as honest gaps, not claimed.
- Where a player genuinely cannot be autonomously driven (secure surface, hard input), it is an operator-attended SKIP-with-reason per §11.4.52 + a tracked item — never a faked PASS.

Non-goals (YAGNI)

- Other platforms (phone/desktop/web) this pass.
- epub if non-trivial (pdf covers the book case for the 93 pdf items).

- Music_artist/song/game/software types (0 or non-playable content; software isn't "played").